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Relational Theory & Concepts

A relation is a mathematical set of tuples that all share the same set of attributes. A relation consists of:

- Attributes (columns)
- Tuples (rows)

In relational theory:

A relation R over attributes

$$A_1, A_2, A_3, \dots, A_n$$

is:

A finite set of tuples, where every tuple contains one value for each attribute, and every value belongs to the attribute's domain.

4. Properties of a Relation

A relation must satisfy several mathematical properties.

Property 1 — Every Relation Has a Unique Name

Example:

- Students
- Departments
- Courses

Each relation in a database must be uniquely identifiable.

Property 2 — Every Attribute Has a Unique Name

Correct:

StudentID	Name	CGPA
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Property 3 — All Tuples Have the Same Attributes

Every row follows the same structure.

Correct:

ID	Name	CGPA
006	aadya	7.97

Every row has exactly three values. i.e id , name , cg , eg 006 , aadya , 7.97 Different rows cannot have different structures within the same relation.

Property 4 — Every Cell Contains a Single (Atomic) Value .

Property 5 — All Values in a Column Come from the Same Domain

Example:

Age column

Property 6 — Duplicate Tuples Are Not Allowed .

Because a relation is a set.

A set never contains duplicate elements.

SQL allows duplicates.

Doesn't that contradict the relational model?

Yes.

SQL is not a pure implementation of relational theory.

Property 7 — Order of Rows Does Not Matter

Without an ORDER BY clause,
SQL does not guarantee row order.
The database may return rows in any sequence.

Property 8 — Order of Columns Does Not Matter , However, in SQL, column order affects how results are displayed and how SELECT * expands, so implementations preserve an order.

Property 9 — A Relation Is Finite

A database relation contains a finite number of tuples . Codd's relational model assumed every attribute had a valid value from its domain.
The treatment of missing information became a subject of later debate , which introduced NULL , supported by sql ,

10. Relation vs SQL Table

Relational Theory	SQL Table
No duplicate tuples	Duplicate rows are allowed unless prevented by constraints or queries
Unordered rows	Row order is not guaranteed unless ORDER BY is used
Unordered attributes (theoretical)	Columns have a defined display order
Pure mathematical set	Practical implementation
Originally did not include SQL-style NULL semantics	Supports NULL values.

Is a relation exactly the same as a table?

No.

A table is the SQL implementation of the relational concept. While they are closely related, SQL tables allow features such as duplicate rows and NULL values that differ from the original mathematical definition of a relation.

SQL is a practical implementation designed for real-world systems. Features such as duplicate rows, NULL values, and implementation-specific behaviors improve usability and flexibility, even though they differ from the original mathematical relational model.